

Geothermal Plant Project (Final Report of

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Code and Dashboard Availability:

The project can be accessed at:

- **Deployed dashboard:** <https://geothermaldashboard-mmrt8zvkhc2nmzrubjpwzh.streamlit.app/>
- **GitHub repository:** <https://github.com/thientruc1691997/geothermaldashboard>

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1 Introduction

Geothermal energy represents the transition to a sustainable, low-carbon power source. In order to produce electricity, geothermal power plants use steam that comes from hot water reservoirs found a few miles below the earth surface. Currently, geothermal electricity generation is used in more than 25 countries around the world. However, like any other energy generating process, it comes with its own set of risks. The operational process, specifically the production and injection of fluids deep into the geological formations of the earth carries an inherent risk of induced seismicity. Although most induced events are micro-seismic and undetectable at the surface, rare high magnitude events can pose safety risks to infrastructure and public acceptance. Thus, for companies, the challenge lies not just in monitoring these events, but in forecasting the seismic events with sufficient time in order to adjust operations proactively to minimize the risk before the event occurs.

In 2019, VITO (Flemish Institute for Technological Research) started their first geothermal energy trial operation just outside Mol, the Balmatt Project. In this study, we use the data (2019-2025) from the Balmatt project to develop a machine learning framework to predict the probability of the occurrence of a seismic event in the next 7 days and the highest magnitude of that event and also to find out which operational variables cause induced seismicity.

2 Exploratory Data Analysis

Due to length constraints in the main report, all exploratory data analysis (EDA) figures and tables are placed in the Appendix. This section provides a narrative overview of the key patterns revealed by those visualizations.

2.1 Data Description

The dataset contains two complementary components spanning the period from 28/11/2018 to 19/09/2025. The operational dataset consists of 695,625 time-stamped measurements across 25 variables describing the state of the geothermal plant, including injection and production conditions, wellhead pressures and temperatures, basin and production flows, energy metrics, and phase-level cumulative quantities (see Table 9). The plant operated in 32 distinct phases, labelled from 0 to 13.3.

In parallel, the seismic-event dataset includes 378 recorded events with 15 variables describing event timing, magnitude, peak ground velocity, epicentral coordinates (x, y, z), distance to the nearest fault, and the operational phase active at the event time (see Table 10).

2.2 Operational Data

2.2.1 Missing Data

The missingness analysis shows that gaps are not randomly distributed. As illustrated in Figure 2a, most operational variables have very low missing rates (typically below 1%). Exceptions include heating-related variables (around 6% missing) and several flow-related variables such as *inj_ap*, *prod_flow*, and *basin_flow* (around 2%). Time-related variables are nearly complete, which is important for chronological modeling. Variables with more than 5% missingness are therefore considered for removal to avoid unstable imputations.

When examining the continuity of the ideal 5-minute sampling interval, however, missingness appears highly structured. Figure 2b shows that missing rows tend to cluster within certain operational phases. Some phases are nearly continuous, while others contain thousands of missing intervals. These blockwise gaps likely correspond to shutdowns, maintenance, or offline monitoring periods. Because the gaps occur in non-random clusters, temporal preprocessing must be applied carefully (e.g., forward-filling, interpolation, and phase-aware feature engineering) to avoid introducing leakage or distorting time-dependent patterns.

2.2.2 Operational Patterns

Beyond missingness, the operational patterns vary significantly across time. As shown in Figure 3, the dataset naturally segments into distinct operational regimes. A large gap appears between 2019 and 2021, likely reflecting extended shutdowns during the COVID-19 period.

Figure 4 highlights variables that consistently drop to zero when a phase ends or when `is_producing = False`, including *inj_flow*, *inj_whp*, *basin_flow*, *prod_flow*, *volume*, *inj_energy*, and *cooling_energy*. These patterns are not physical but reflect sensor or system behavior during non-producing periods.

2.3 Seismic Event Data

Across the full monitoring period, nearly 400 seismic events were recorded (Figure 6). The magnitude distribution is highly imbalanced: only two events exceed $M \geq 1.8$, and six fall within $1.2 \leq M < 1.8$. More than 98% of events lie below magnitude 1.2, which poses challenges for modeling rare high-magnitude events.

Seismicity also exhibits strong temporal clustering. As shown in Figure 7, multiple events often occur within the same minute or within short time windows. These bursts may span multiple magnitude categories, complicating attempts to distinguish event severity based on short-term operational changes.

3 Data Preprocessing

3.1 Resampling and Cleaning

Operational timestamps (`recorded_at`) were converted to datetime format and the data were sorted by phase and time to preserve chronological order. Variables with more than 5% missingness (`hedh_thpwr`, `heat_exch_energy`, `cum_heat_exch_energy`) were removed from the feature set.

The remaining operational data were then resampled to a uniform 5-minute grid, increasing the dataset to 716,312 rows. Figure 5 illustrates the raw versus cleaned trajectories: the raw signals contain gaps and irregular sampling, while the cleaned versions show consistent time steps with imputed values marked in red.

Within each operational phase, time gaps between consecutive records were computed and classified as short (≤ 15 minutes) or long (> 15 minutes). Missing values were imputed separately within each phase and only for producing periods (`is_producing = True`). Short gaps were filled using time interpolation to obtain smooth trajectories. For long gaps, single missing rows were imputed using Multiple Imputation by Chained Equations (MICE), while longer sequences were filled using a combination of forward fill within the phase.

Finally, `prod_flow` was recomputed as `inj_flow - basin_flow`, and some cumulative variables (`cum*`) were recalculated as phase-wise cumulative sums of their corresponding base variables.

3.2 Handling Zero Segments

During resampling, upsampling, and imputation, the variables that systematically drop to zero in non-producing periods (`inj_flow`, `inj_whp`, `basin_flow`, `prod_flow`, `volume`, `inj_energy`, `cooling_energy`) are explicitly set to zero to maintain consistency and avoid introducing artificial operational signals. After imputation, these zero constraints are re-enforced for all non-producing rows.

3.3 Merging Seismic and Operational Data

For each operational timestamp t , all seismic events with timestamps in the interval $(t, t + 7 \text{ days}]$ are associated with the corresponding operational record. Using this merged timeline, three prediction targets are defined for each t : (i) a binary label, (ii) a multi-class label, and (iii) the maximum event magnitude in the next 7 days. These targets are obtained by scanning the 7-day forward window $(t, t + 7 \text{ days}]$ and assigning values based on event occurrence and the maximum observed magnitude.

4 Modeling Approach

This section summarises the modelling strategy used in this project. The objectives are: (i) to predict the probability that at least one seismic event will occur in the next seven days, and (ii) to estimate its expected strength. The target is grouped into four classes:

- **Class 1:** no event,
- **Class 2:** event with magnitude < 1.2 ,
- **Class 3:** event with magnitude ≥ 1.2 and < 1.8 ,
- **Class 4:** event with magnitude ≥ 1.8 .

A secondary aim is to identify which operational variables are most related to induced seismicity.

4.1 Initial Experiments and Data Leakage

The first attempts used Logistic Regression and Random Forest classifiers with a standard random train–test split. Logistic Regression served as a simple baseline, and Random Forests were used to capture nonlinear links between operational variables, seismic history, and event occurrence.

However, because the data is time series, the random split introduced *data leakage*: future samples were included in the training set while the model was tested on earlier timestamps. This produced unrealistically high scores. Since the setup violated the temporal structure of the data, all random-split results were discarded. From this point onwards, every experiment used strict temporal splits.

4.2 Two-stage model

Our data are strongly imbalanced: most days have no seismic event, and only a small number contain an event. Because of this, training a single model to predict magnitude classes did not work well. To make the task easier, we use a two-stage model. In the first stage, a *Gatekeeper* checks whether a seismic event will happen or not. If an event is predicted, the second-stage *Expert* model estimates how strong the event may be. This approach reduces the impact of imbalance and makes the prediction process more stable and intuitive.

4.2.1 Stage 1 (Gatekeeper): Event-Probability Forecasting with Temporal Splits

After identifying leakage, the pipeline was rebuilt using *temporal* train–validation splits. Models are always trained on older data and tested on later periods. To further prevent leakage, a **7-day gap** is placed between the end of the training window and the start of the validation window, matching the 7-day forecast horizon.

Model selection uses **5-fold time-series cross-validation** with an expanding training window. Each fold trains on all data up to a certain point, leaves a 7-day gap, and validates on the following block. The validation window moves forward in time, while the training set grows, ensuring clean temporal separation in every fold.

Random Forests remained the main model for predicting event probability. A first design used a global two-stage *Gatekeeper–Expert*, where the Gatekeeper predicts whether an event will happen and the Expert predicts its magnitude bin. However, the global Gatekeeper behaved inconsistently across operating regimes, and threshold tuning based on the Precision–Recall curve led to unrealistically small thresholds.

To better handle changing operating conditions, the final model uses a Mixture-of-Experts (MoE) Gatekeeper. Operating states are identified with K-means clustering using three physically meaningful variables: net flow, the temperature difference $\Delta T_{\text{prod-inj}}$, and the injection wellhead pressure (inj_ap). These variables are selected because previous studies have shown that they are strongly linked to induced seismicity.

The variable `net_flow` was extracted to quantify the imbalance between injection and production, as such imbalances are known to alter pore pressure and stress conditions in the subsurface, thereby increasing the likelihood of induced seismicity [1]. The temperature difference $\Delta T_{\text{prod-inj}}$ influences thermoelastic stresses in

the rock; large temperature contrasts can create contraction or expansion that perturbs the local stress field and may trigger seismic events [2]. Injection pressure (*inj_ap*) represents the direct hydraulic forcing applied to the reservoir, and higher injection pressures have been repeatedly associated with elevated seismicity rates in geothermal and wastewater injection operations [3].

Using these three variables for clustering allows the K-means algorithm to group time periods with similar hydraulic and thermal loading conditions. A separate Random Forest expert is then trained for each operating-state cluster, while a global model serves as a fallback for clusters with very few samples.

A separate Random Forest expert is trained for each cluster, and a global model is used for small clusters. This MoE Gatekeeper achieved the best performance for Objective 1, judged by the F1-score for the event class and by a more realistic balance between missed events and false alarms.

In the binary setting, let TP , FP , and FN denote true positives, false positives, and false negatives. Precision and recall are

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}.$$

The F1-score is

$$F1 = \frac{2 \text{precision recall}}{\text{precision} + \text{recall}}.$$

This score is the main metric for event-probability forecasting.

4.2.2 Stage 2 (Expert): Magnitude Forecasting

The second objective is to predict the maximum magnitude in the upcoming seven days. The task was first treated as a multi-class classification problem using magnitude bins, but large events were extremely rare, and the classifier performed poorly for the highest-magnitude classes.

The problem was therefore reformulated as a regression task. A Random Forest regressor predicts the continuous maximum magnitude, but only for timestamps where the Gatekeeper predicts that an event will occur. This avoids arbitrary bin definitions and makes full use of the magnitude values. The same 5-fold expanding-window cross-validation with a 7-day gap is applied.

Model performance is evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}.$$

MAE measures the average prediction error, and RMSE penalises large errors more strongly. Under temporal validation, the Random Forest regressor provided stable and accurate predictions and was therefore selected as the final model for Objective 2.

4.3 Handling Class Imbalance

Only a small fraction of timestamps contain events, so class imbalance is a key issue. In the Gatekeeper, we use *class weights* based on class frequencies in the training portion of each fold. These weights increase the importance of rare event samples while avoiding leakage, because they are computed strictly within the training set.

SMOTE oversampling was also tested, applied only to the training data after defining the temporal split and the 7-day gap. Although SMOTE sometimes improved recall, it altered the data distribution in early operating regimes, so it was not used in the final models.

In the MoE design, each operating-state cluster has its own expert. This naturally reduces imbalance inside clusters. In Stage 2, the regression model is trained only on timestamps with events, so imbalance is no longer a problem.

4.4 Final Feature Selection and Model Pipeline

All final models use temporal splits, expanding-window cross-validation, and a 7-day gap. Preprocessing includes aligning operational and seismic data, resampling to a common time resolution, and imputing missing values using interpolation, MICE, forward fill when needed. Change-point detection with binary segmentation is used to identify regime shifts, and models are trained mainly on more stable periods.

The feature set comes from two sources: the operational time series and the seismic catalogue. From the operational data, we extract variables such as flows, pressures, temperatures, cumulative injected volumes, and the production indicator. From the seismic data, we use event times and magnitudes.

From these bases, we derive additional features such as net flow and the temperature difference $\Delta T_{\text{prod-inj}}$, as well as seismic event counts in different magnitude ranges over 1-, 3-, and 7-day windows. Rolling statistics (mean, max, and standard deviation) over the same windows are computed for selected variables to capture short- and medium-term changes in operational conditions and seismic activity.

These features were retained because they are physically meaningful and frequently identified as important in feature-importance and permutation analyses. The resulting pipeline corrects the weaknesses of the initial random-split experiments and provides a methodology that is both statistically reliable and geomechanically interpretable.

The full pipeline is shown as Figure ?? in the Appendix.

4.5 Feature Importance Analysis

Feature importance was evaluated on the Stage-1 gatekeeper Random Forest classifier. For each fold of the time-series cross-validation, the model was trained on the temporally earlier data (with a 7-day gap before the test window) using RandomOverSampler to balance the binary event label. After training, we extracted

the impurity-based importance of each predictor, defined as the mean decrease in Gini impurity attributed to that feature across all trees in the ensemble.

To obtain a robust ranking, the importance scores were computed separately in every cross-validation fold and then averaged across folds. The values reported in the main results section correspond to this cross-validated mean importance. Because impurity-based importance can be diluted among strongly correlated predictors, these numbers are interpreted as a relative ranking of the most influential variables rather than as absolute physical sensitivities.

5 Main Results

5.1 Initial Experiments and Data Leakage

The problem was initially formulated as a multi-class task predicting magnitude bins. Early experiments (Table 1) showed seemingly high performance for both Logistic Regression and Random Forest, even without extensive tuning. However, these results were obtained under a random train–test split, which ignores temporal ordering. As a consequence, future observations leaked into the training set, producing artificially inflated scores and severe overfitting.

Table 1: Initial (leakage-affected) results using random train–test splits for Logistic Regression and Random Forest.

Model	Variant	Test Accuracy	F1 Macro
Logistic Regression			
	Original	0.8951	0.6500
	SMOTE	0.7866	0.7824
	Undersampling	0.7839	0.7791
Random Forest			
	Original	0.9964	0.9961
	SMOTE	0.9964	0.9962
	Undersampling	0.9932	0.9936

Table 2: Test set confusion matrices obtained under random train–test splits (leakage-affected).

(a) Logistic Regression (SMOTE)					(b) Random Forest (SMOTE)				
	C1	C2	C3	C4		C1	C2	C3	C4
C1	49 957	5 879	3 120	44	C1	58 737	262	0	0
C2	2 217	6 576	1 765	571	C2	6	11 121	1	0
C3	217	97	786	0	C3	0	3	1 098	0
C4	0	0	0	403	C4	0	1	0	402

5.2 Two-stage model

5.2.1 Stage 1 (Gatekeeper): Event-Probability Forecasting

The results in Table 3 show that the Gatekeeper’s macro-F1 changes a lot across folds, with an average of only 0.41. This means the binary model still has difficulty separating “event” from “no event”, even after using the optimal threshold for event detection.

The confusion matrix in Table 4 also shows many false positives and false negatives, with a large number of missed events (5584). This directly limits the performance of the second-stage magnitude model.

 Table 3: Stage 1 Gatekeeper – $F1_{\text{macro}}$ per fold

Fold	$F1_{\text{macro}}$
1	0.000
2	0.648
3	0.797
4	0.033
5	0.577
Mean	0.411
Std	0.330

Table 4: Stage 1 Gatekeeper – Confusion matrix (final fold)

Actual	Predicted	
	0: No Event	1: Event
0: No Event	98706	11054
1: Event	5584	2505

To improve this, we replaced the single Gatekeeper with a Mixture-of-Experts (MoE), so that different experts can focus on different operating regimes and reduce the impact of data imbalance and non-linear behaviour.

5.2.2 Stage 1 (Gatekeeper): Event-Probability Forecasting with Mixture of Expert (MoE)

The MoE Gatekeeper clearly improves the binary event detection: the average macro-F1 increases from about 0.41 to about 0.56 and is more consistent across folds. The final-fold confusion matrix in Table 6 shows many more true events are correctly detected (1466 vs 575 missed), at the cost of extra false positives, which is an acceptable trade-off for a gate whose main goal is not to miss real events. In addition, the MoE Gatekeeper

uses regime detection to remove the unstable initial operating period (regimes 0–1), so Stage 1 of the MoE model is trained and evaluated only on data from regime 2 onward. As a result, the number of “event” and “no-event” samples differs from the earlier Gatekeeper. The detailed regime segmentation is shown in Figure 9 in the Appendix.

Table 5: Stage 1 MoE Gatekeeper – $F1_{\text{macro}}$ per fold (binary)

Fold	$F1_{\text{macro}}$
1	0.530
2	0.485
3	0.597
4	0.478
5	0.715
Mean	0.561
Std	0.088

Table 6: Stage 1 MoE Gatekeeper – Confusion matrix (final fold, thresh = 0.30)

Actual	Predicted	
	0: No Event	1: Event
0: No Event	44918	2791
1: Event	575	1466

5.2.3 Stage 2 (Expert): Multiclass prediction

Stage 2 is implemented as a multi-class classifier that aims to distinguish between not-severe and severe events, but it is only applied to the samples that the Stage 1 MoE Gatekeeper has predicted as events. For the final fold, this corresponds to 1466 detected events. The severity labels are grouped into two classes: class 1 (*not severe*) and class 2 (aggregating the original classes 2–3 as *severe*). The resulting confusion matrix is shown in Table 7.

Table 7: Stage 2 Expert – Confusion matrix for severity classification on detected events (final fold).

Actual severity	Predicted severity	
	1: Not severe	2: Severe
1: Not severe	25	0
2: Severe	1441	0

The classifier assigns all detected events to the not-severe class, correctly labeling only 25 out of 25 not-severe events but failing to identify any of the 1441 severe events. This leads to an overall accuracy of about 2% and a macro F1-score of approximately 0.02, with the severe class having zero precision, recall, and F1. These results indicate that the multi-class Stage 2 expert is not reliable under the current data imbalance. For this reason, the subsequent analysis focuses on a regression formulation of Stage 2, where the model predicts continuous magnitude rather than discrete severity classes.

5.2.4 Stage 2 (Expert): Regression prediction

The regression performance of Stage 2 in Table 8 is reasonably good in the first three folds, with RMSE values below 0.30. However, RMSE increases substantially in folds 4 and 5, reaching as high as 1.17. Given that the target magnitude values lie roughly within the range (-1.0, 2.0), these errors are relatively large. This suggests that the later part of the time series contains more irregular operational regimes or rarer event patterns that the model struggles to generalize to.

Table 8: Stage 2 (Expert) Regression Performance Across Folds

Fold	MAE	RMSE
Fold 1	0.208	0.253
Fold 2	0.400	0.400
Fold 3	0.255	0.260
Fold 4	0.591	0.742
Fold 5	1.169	1.173
Average	0.525	0.565

5.3 Feature Importance and Physical Interpretation

Figure 1 reports the top 20 predictors for the Stage 1 Random Forest Gatekeeper, ranked by cross-validated mean impurity-based importance. The model assigns highest importance to short-term production well-head pressure statistics (`prod_whp_mean_1d`, `prod_whp_max_1d`, and related features) and to injection annulus pressure descriptors (`inj_ap_mean_1d`, `inj_ap_max_7d`). Cumulative injection and cooling energy and cumulative volume are also highly ranked, together with recent counts of larger events over 3- and 7-day windows (`cnt_M1t12_3d` and `cnt_M1t12_7d`). This pattern is consistent with physical intuition: short-term changes in pressure, flow and temperature—combined with recent seismic response—appear to control short-term fault susceptibility. From VITO’s perspective, the feature importance analysis therefore supports the view that induced seismicity in this reservoir is tightly coupled to recent operational history on time scales of a few days.

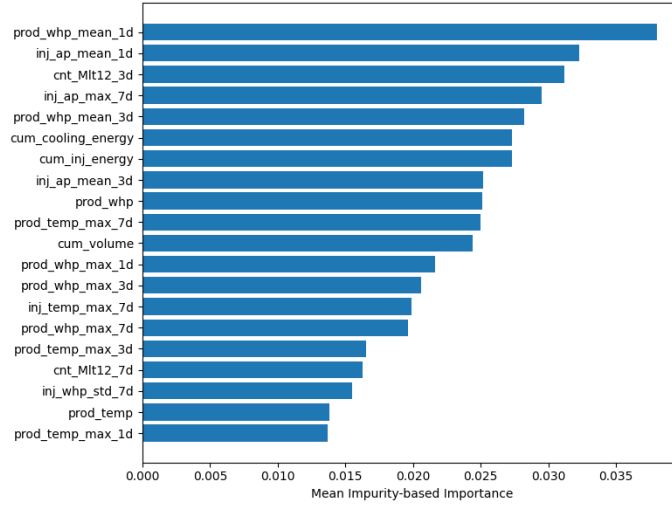


Figure 1: Top 20 features for the Stage-1 MoE gatekeeper, ranked by mean impurity-based importance (Gini) over all time-series CV folds.

5.4 Limitations

The modelling results should be interpreted with several limitations in mind. First, the dataset is highly imbalanced, with very few medium or large events; this restricts the model’s ability to learn reliable patterns for higher-severity seismicity. Second, the regression performance degrades in the later folds, indicating that regime shifts or changes in operational behaviour are not fully captured by the current feature set. Third, because the model produces an independent prediction every five minutes, the resulting probability curve may fluctuate noticeably over time, introducing temporal noise that does not necessarily reflect true changes in the subsurface conditions. Finally, the MoE approach depends on the quality of the regime segmentation; if the clustering does not align with real operational states, expert models may underperform.

5.5 Practical Interpretation for VITO

Based on the MoE results, the event–detection stage achieves an estimated reliability of roughly 60%, meaning the model correctly identifies about two–thirds of the events that actually occur. This level of sensitivity is useful for early warning, but it can be increased further by lowering the decision threshold (currently set at 30%). Reducing the threshold would allow the model to detect more events, though at the cost of producing more false alarms. The appropriate operating point therefore depends on the company’s tolerance for missed events versus false positives.

In practical terms, the model takes the current operational state (measured every five minutes) and predicts the probability that at least one induced seismic event will occur in the next seven days. As a result, a seven-day event probability is available at each five-minute time step, which can be visualised as a continuous risk curve over time in the next 7 days.

6 Recommendation

The results of this work point to four main directions for VITO. First, the most critical bottleneck is the lack of data for rare high-magnitude events: while the model performs reasonably well when sufficient observations are available (e.g. for events with $M < 1.2$), it cannot learn meaningful decision boundaries for regimes with $M \geq 1.2$, where only around 8 events are observed. Targeted efforts to collect, harmonise, or pool additional data for these rare classes are therefore essential. Second, the current modelling framework can serve as a strong baseline for exploring more advanced deep learning architectures, which may better capture nonlinear temporal dependencies and interactions between operational variables and seismicity. Third, to maximise practical impact, the modelling workflow should be converted into a production-ready pipeline, exposed via APIs and integrated with the operational monitoring system so that forecasts can be generated, logged, and monitored in real time. Finally, because the data distribution drifts across operating regimes, an adaptive or continual learning strategy is required: periodically updating the training set with new data and retraining the model through the pipeline would help keep the deployed model calibrated and robust as the geothermal system evolves.

7 Conclusion

This project developed a time-aware modelling framework to forecast short-term induced seismicity at the Balmatt geothermal site. After correcting data leakage and using expanding-window cross-validation with a 7-day gap, the binary Mixture-of-Experts Gatekeeper emerged as the most reliable component for practical use. It detects a meaningful proportion of upcoming events and provides a robust *relative risk ranking* of operational periods, which can support VITO in identifying intervals with elevated seismic sensitivity. When the Gatekeeper signals an event, the Stage 2 Random Forest regressor offers a coarse estimate of the expected maximum magnitude within the 7-day horizon, sufficient for prioritising scenarios but not for safety-critical prediction.

The main limitations arise from the extreme imbalance in the seismic catalogue and the scarcity of larger events, which restrict the accuracy of magnitude forecasts and prevent calibrated probability interpretation. The model should be applied only within the range of historical operating conditions and not extrapolated to fundamentally new regimes.

Reference

- [1] National Research Council. *Induced Seismicity Potential in Energy Technologies*. Washington, DC: The National Academies Press, 2013.
- [2] Jon Rutqvist. “A review of coupled thermal–hydrological–mechanical processes in geothermal systems”. In: *Geothermics* (2015).
- [3] Arthur McGarr. “Maximum magnitude earthquakes induced by fluid injection”. In: *Journal of Geophysical Research: Solid Earth* (2014).

Appendix: Figures, Tables & Pipeline

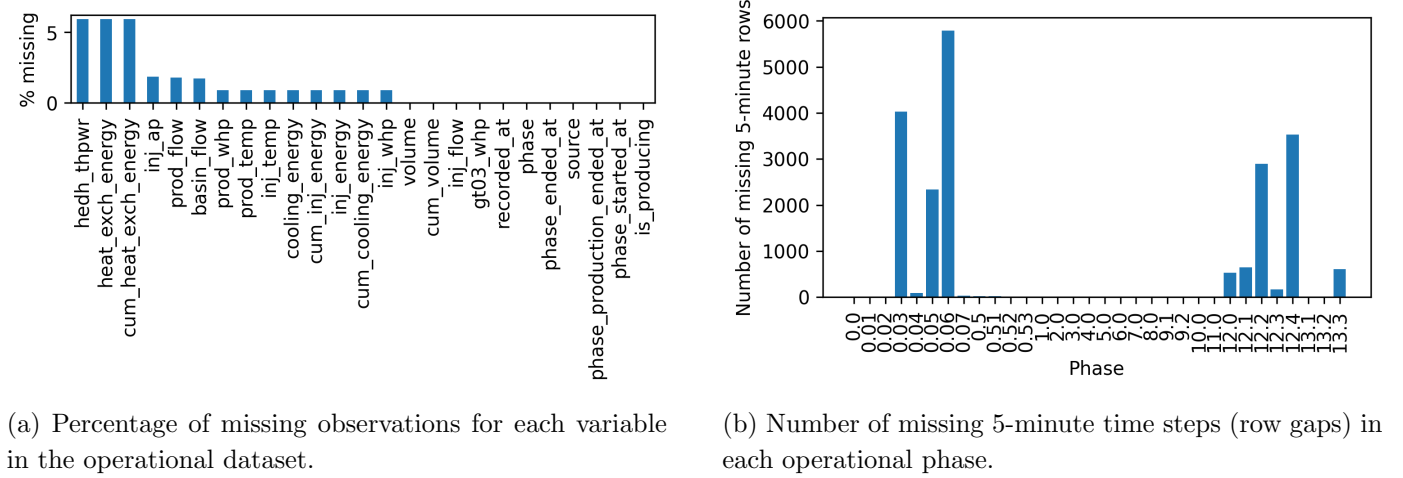


Figure 2: Overview of missing data patterns in the operational dataset: (a) variable-wise missingness; (b) row-wise missingness within phases.

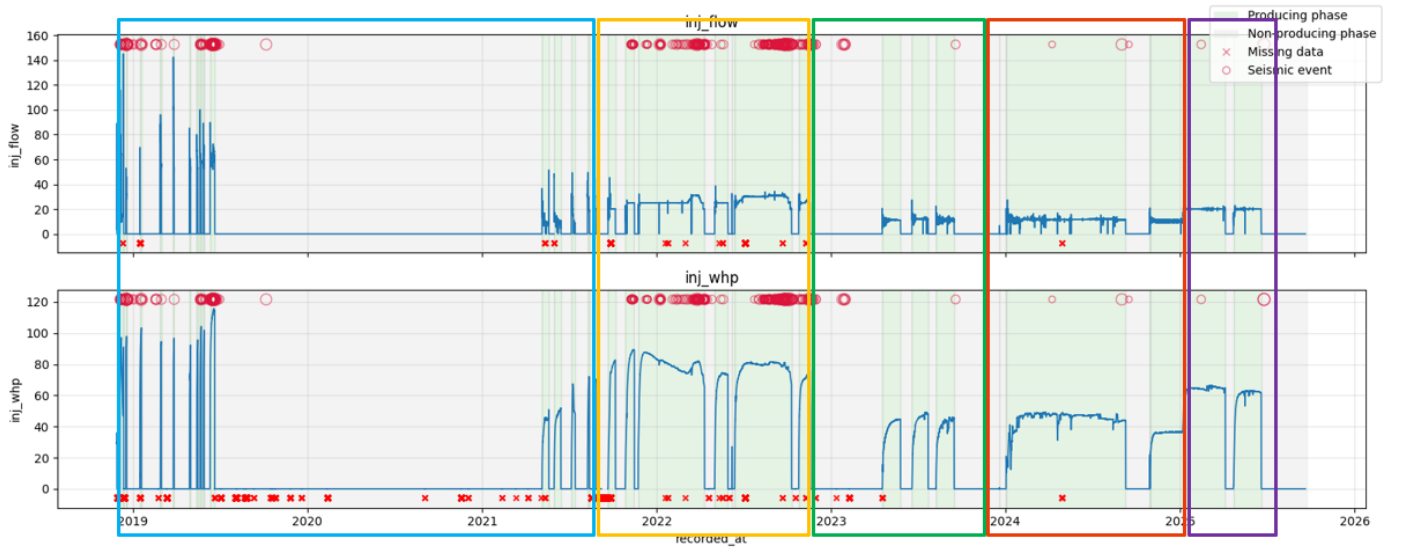


Figure 3: Time series plot of selected operational variables, illustrating distinct clusters that reflect different activity regimes throughout the plant's lifetime. The segmented periods highlight transitions between operational modes, shutdown intervals, and changes in system behavior, providing a visual indication of how operational patterns evolve across time.

Table 9: Variables in the operational metrics dataset

Variable	Description	Type
Recorded_at	Timestamp of the operational record	Datetime
Phase	Operational phase identifier	Categorical
Inj_flow	Total injection flow rate [m ³ /h]	Numeric
Inj_whp	Injection wellhead pressure [bar]	Numeric
Inj_temp	Injection wellhead temperature [°C]	Numeric
Inj_ap	Injection annular pressure [bar]	Numeric
Prod_temp	Production wellhead temperature [°C]	Numeric
Prod_whp	Production wellhead pressure [bar]	Numeric
Gt03_whp	Wellhead pressure in GT03 well [bar]	Numeric
hedh_thpwr	Thermal power at the district-heating heat exchanger [kW]	Numeric
basin_flow	Basin injection flow rate [m ³ /h]	Numeric
prod_flow	Production flow rate [m ³ /h] (inj_flow – basin_flow)	Numeric
is_producing	Indicator whether the plant is producing energy	Boolean
phase_started_at	Start time of the operational phase	Datetime
phase_production_ended_at	End time of production in this phase	Datetime
Phase_ended_at	End time of the operational phase	Datetime
volume	Injected volume during the record [m ³]	Numeric
cum_volume	Cumulative injected volume in this phase [m ³]	Numeric
inj_energy	Injected energy [MWh]	Numeric
cum_inj_energy	Cumulative injected energy in this phase [MWh]	Numeric
cooling_energy	Energy extracted from brine before reinjection [MWh]	Numeric
cum_cooling_energy	Cumulative extracted energy in this phase [MWh]	Numeric
heat_exch_energy	Heat-exchanger energy on district-heating side [MWh]	Numeric
cum_heat_exch_energy	Cumulative heat-exchanger energy in this phase [MWh]	Numeric

Table 10: Variables in the seismic events dataset

Variable	Description	Type
occurred_at	Timestamp of the seismic event	Datetime
event_id	Unique identifier of the seismic event	Numeric
phase	Operational phase during which the event occurred	Categorical
is_producing	Indicator whether the plant was producing at the event time	Boolean
phase_started_at	Start time of the corresponding operational phase	Datetime
phase_production_ended_at	End of production in the corresponding phase	Datetime
phase_ended_at	End of the corresponding operational phase	Datetime
pgv_max	Maximum peak ground velocity [m/s]	Numeric
magnitude	Local magnitude of the event	Numeric
Hourly_seismicity_rate	Number of events with $M > 0.5$ in the last hour	Numeric
Distance_to_fault	Distance from epicenter to nearest mapped fault	Numeric
Location	Epicentral coordinates in EPSG 31370 (Lambert 72)	Character
x	Epicenter x-coordinate (easting)	Numeric
y	Epicenter y-coordinate (northing)	Numeric
z	Epicenter depth coordinate	Numeric

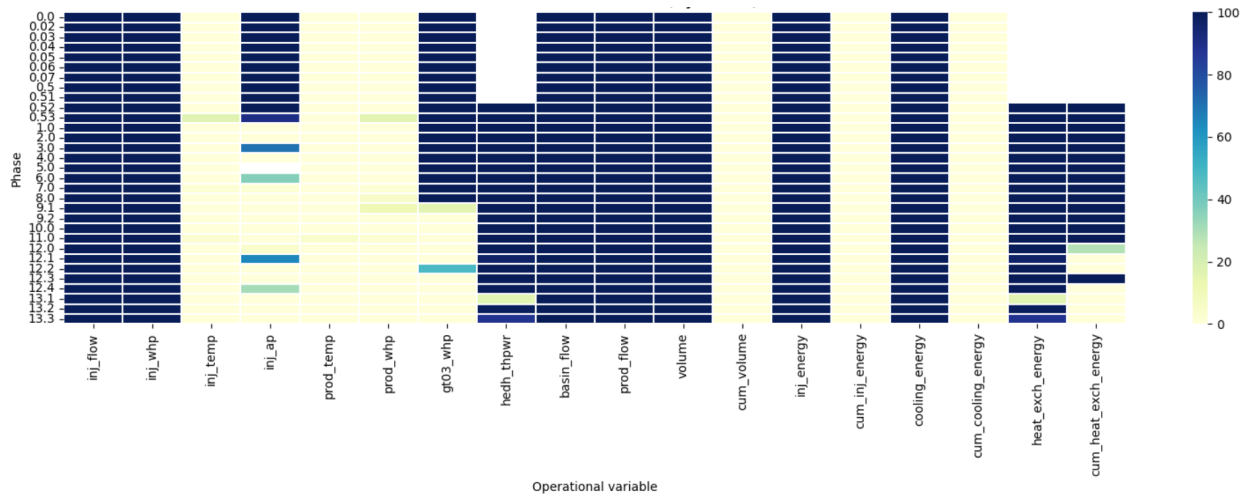


Figure 4: Proportion of zero values for each operational variable across all phases

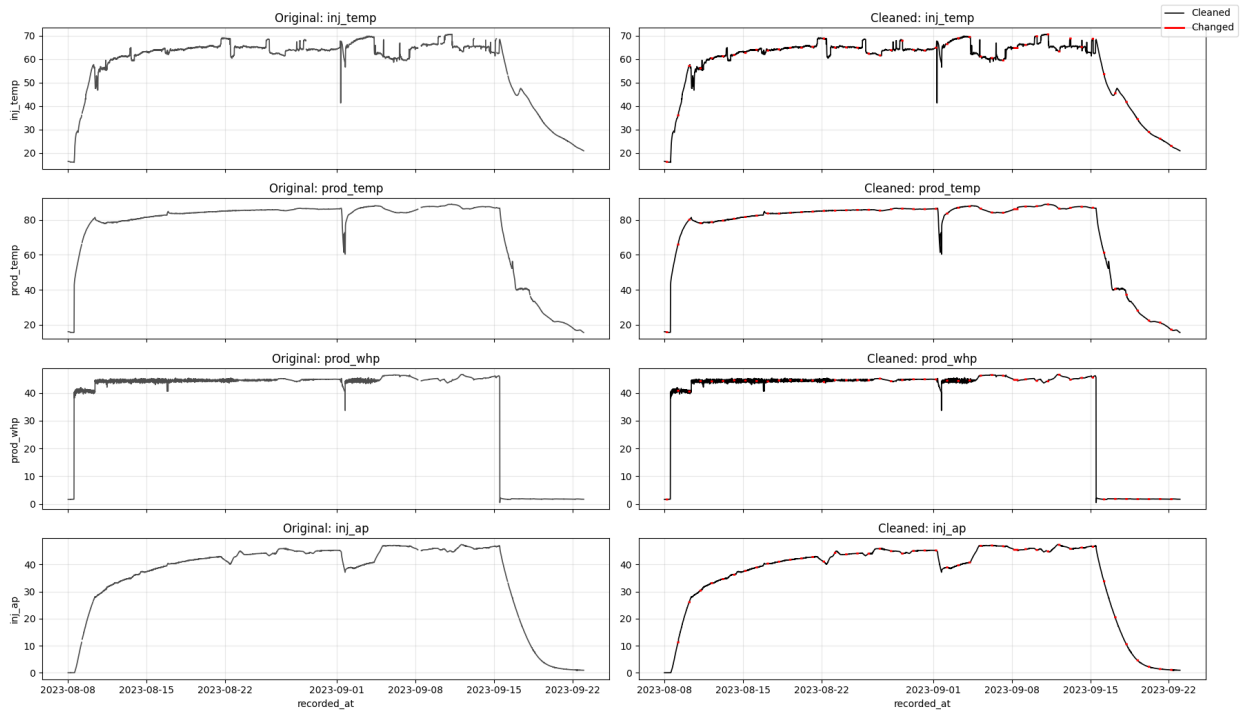


Figure 5: Overview of selected features before (left) and after (right) resampling and cleaning. Red markers indicate imputed values.

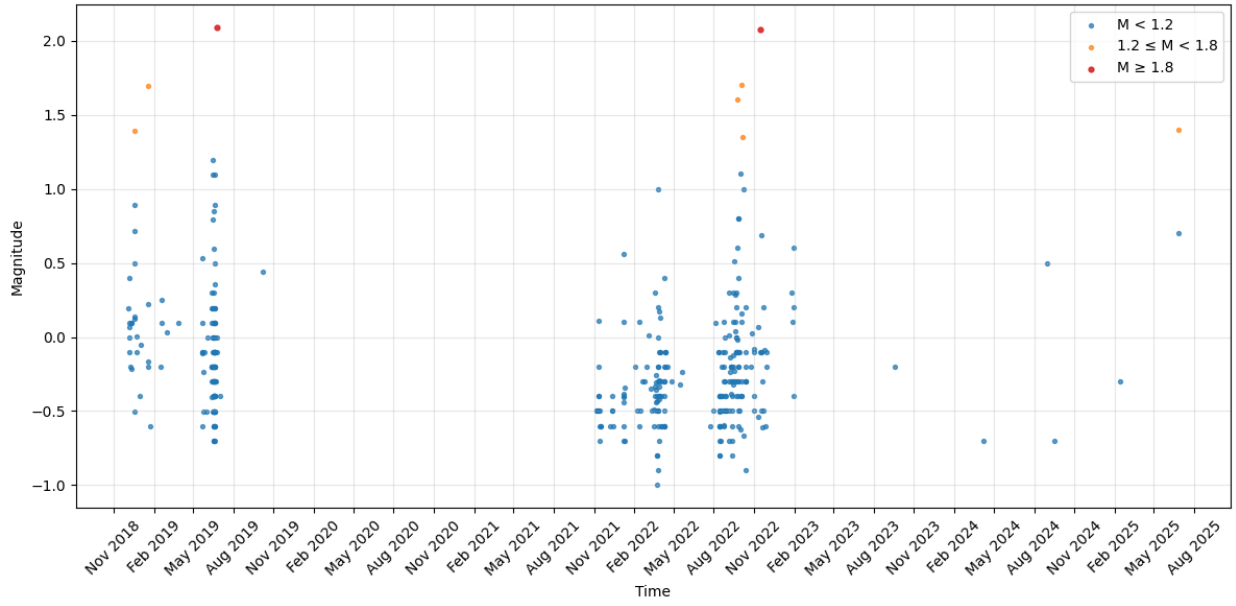


Figure 6: Time-magnitude distribution of all recorded seismic events from 2018 to 2025. Most events fall below magnitude 1.2, while only a small number of moderate and large events occur sporadically. The plot also reveals distinct periods of clustered seismicity separated by long intervals of little to no activity.

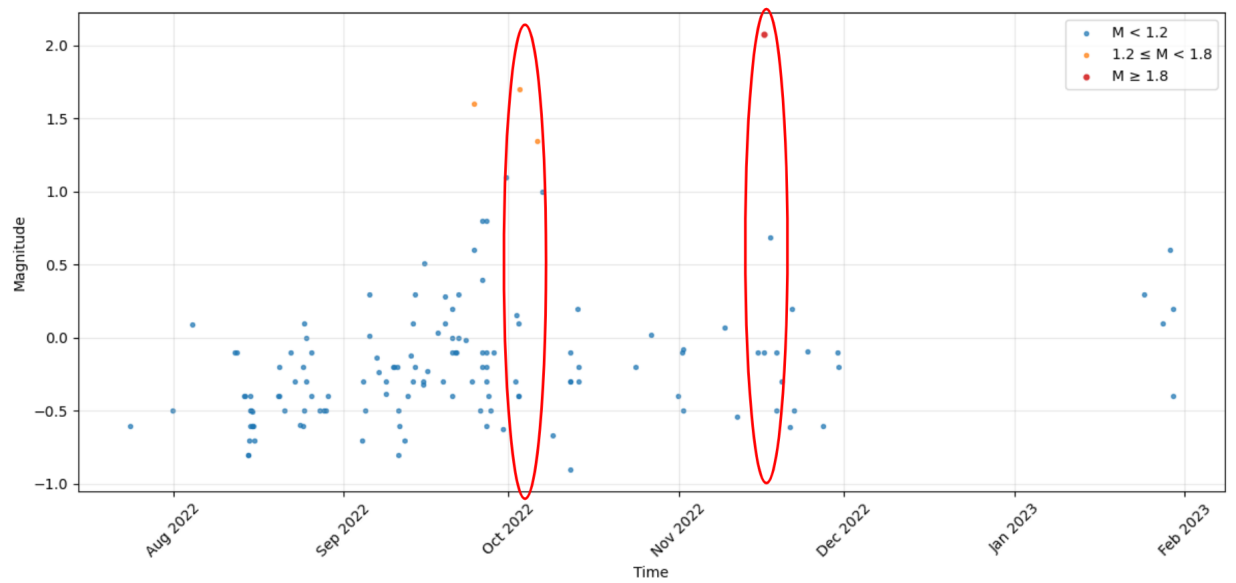


Figure 7: Seismic events between August 2022 and February 2023. The highlighted clusters (red ellipses) mark two short periods in which higher-magnitude events occurred, indicating temporary increases in seismic activity compared to the surrounding months.

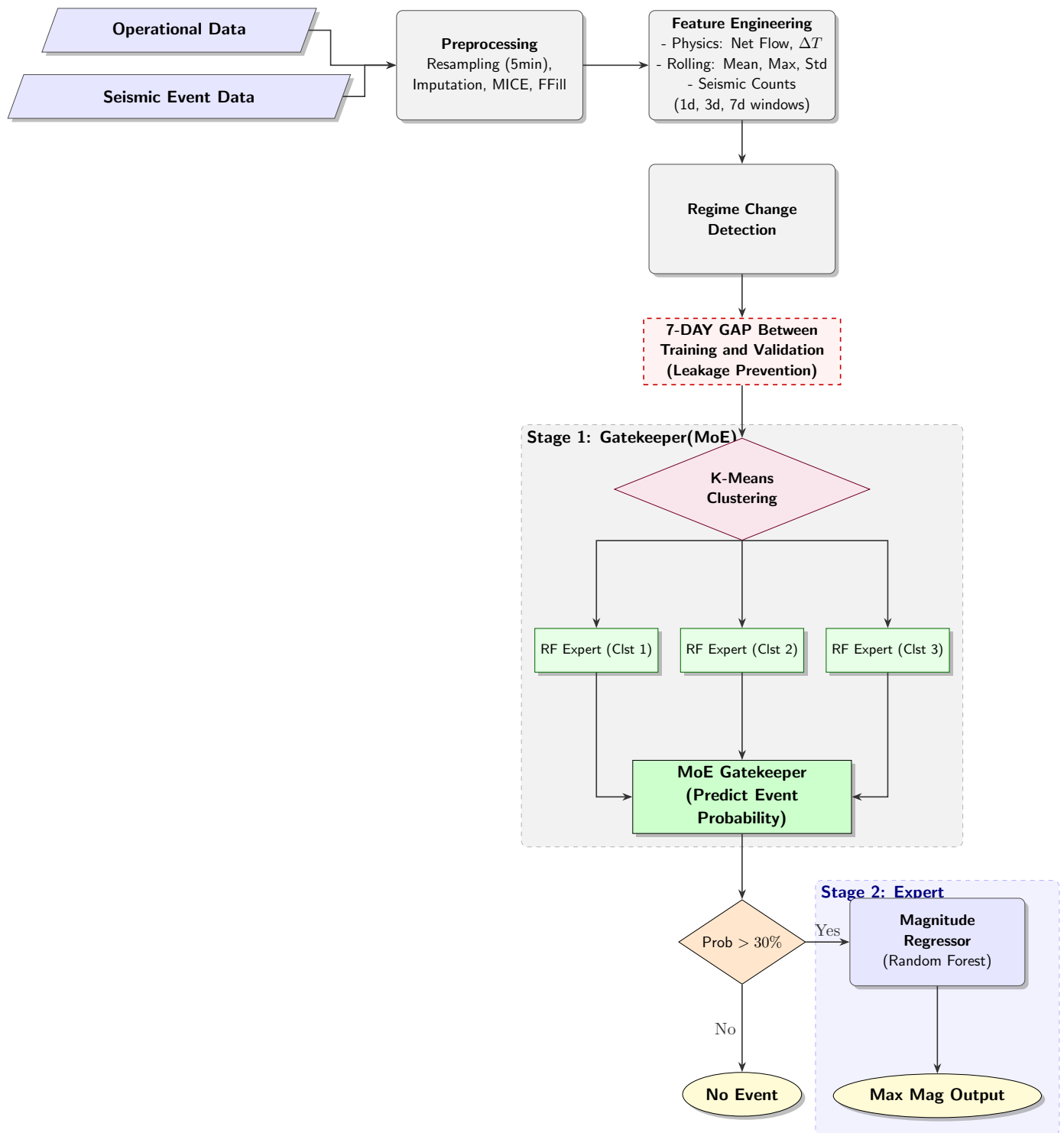


Figure 8: Final Model Pipeline

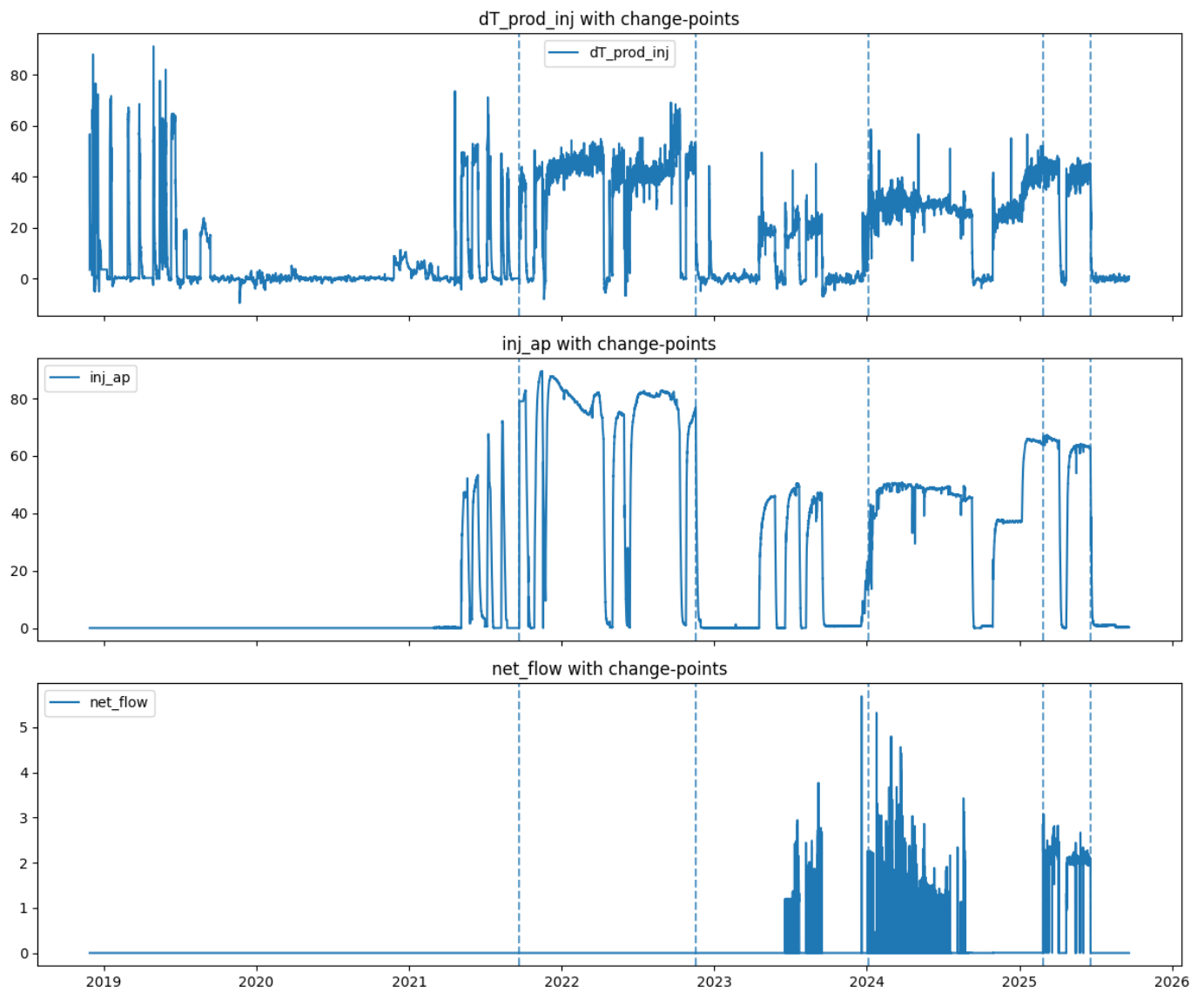


Figure 9: Detect 5 regime changes